

# Kambhammettu Madhu Sudhan Rao

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linkedin.com/in/madhuu-k | github.com/Madhuu-k | Portfolio

## PROFESSIONAL SUMMARY

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Computer Science undergraduate specializing in real-time Computer Vision and Deep Learning. Built AI pipelines, including pose-estimation models achieving 30+ FPS on CPU and AR systems with 40% reduced visual jitter. Seeking an ML/SDE internship focused on real-time computer vision and efficient ML systems.

## TECHNICAL SKILLS

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**Programming Languages:** Python, C++, Java, JavaScript, SQL

**Machine Learning:** TensorFlow, Scikit-learn, Deep Learning, CNNs, Transfer Learning, Model Training, Model Evaluation

**Computer Vision:** OpenCV, MediaPipe

**Web Development:** React, Flask, REST APIs

**Tools:** Git, GitHub, Docker

## PROJECT EXPERIENCE

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- AI Fitness Trainer – Push-Up Counter** | *Python, OpenCV, MediaPipe* | **GitHub** Jan 2026
- Designed a real-time pose estimation pipeline using MediaPipe landmarks to compute joint angles for push-up detection, achieving 30+ FPS on CPU.
  - Implemented a finite-state machine to classify push-up phases while handling occlusion and lighting variations.
  - Evaluated on 1,000+ webcam frames, achieving 96% repetition detection accuracy.
- Plant Leaf Disease Prediction** | *Python, TensorFlow, CNN* | **GitHub** Jan 2026
- Developed a CNN-based image classifier using transfer learning on a dataset of 5,000+ plant leaf images across five disease classes.
  - Applied preprocessing, data augmentation, and stratified dataset splitting to improve model generalization.
  - Achieved 94% validation accuracy across five disease categories.
- Sudarshan Real-Time AR System** | *Python, OpenCV, MediaPipe* | **GitHub** Feb 2026
- Built a markerless AR system using MediaPipe hand landmark tracking to anchor virtual overlays to real-time hand motion.
  - Applied temporal smoothing to stabilize landmark coordinates during dynamic hand movement.
  - Reduced visual jitter by 40% while maintaining end-to-end latency below 100 ms.

## EDUCATION

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### KL University

*Bachelor of Technology in Computer Science, CGPA: 9.4/10*

### Sri Chaitanya Junior College

*Intermediate (MPC), GPA: 9.3/10*

Guntur, India

2023 – 2027 (*Expected*)

Vijayawada, India

2021 – 2023

## RELEVANT CERTIFICATIONS

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- **Oracle Cloud Infrastructure 2025 Generative AI Professional** — Oracle
- **MongoDB Associate Developer** — MongoDB University
- **Python Programming Certificate** — Udemy

## CODING PROFILES

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**LeetCode:** Solved 200+ DSA problems — <https://leetcode.com/u/klu2300031210/>

**GeeksforGeeks:** <https://www.geeksforgeeks.org/profile/madhusudtxfu>